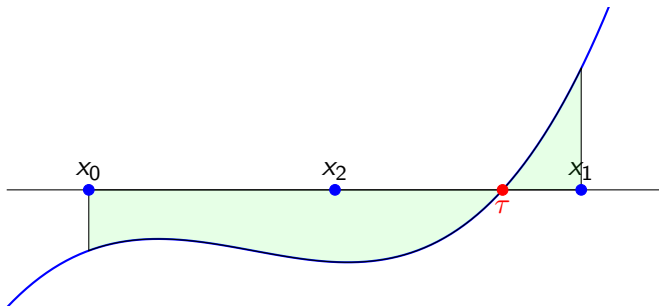


Solving nonlinear equations
1.2: Interval Bisection

MA385 – Numerical Analysis

September 2026



0. Outline

- 1 Bisection
- 2 The bisection method works
- 3 Improving upon bisection
- 4 Exercises

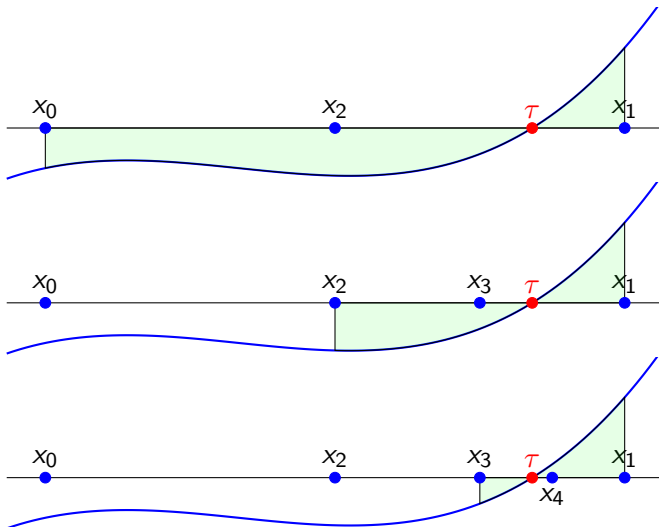
For more details, see Section 1.6 (The Bisection Method) of [Süli and Mayers, *An Introduction to Numerical Analysis*](#)

1. Bisection

The most elementary algorithm is the “*Bisection Method*” (also known as “Interval Bisection”). Suppose that we know that f changes sign on the interval $[a, b] = [x_0, x_1]$ and, thus, $f(x) = 0$ has a solution, τ , in $[a, b]$. Proceed as follows

1. Set x_2 to be the midpoint of the interval $[x_0, x_1]$.
2. Choose one of the sub-intervals $[x_0, x_2]$ and $[x_2, x_1]$ where f change sign;
3. Repeat Steps 1–2 on that sub-interval, until f is sufficiently small at the end points of the interval.

1. Bisection



1. Bisection

This may be expressed more precisely using some *pseudocode*.

The Bisection Algorithm

Set ϵ to be the stopping criterion.

If $|f(a)| \leq \epsilon$, return a . Exit.

If $|f(b)| \leq \epsilon$, return b . Exit.

Set $x_L = a$ and $x_R = b$.

Set $k = 1$

while($|f(x_k)| > \epsilon$)

$x_{k+1} = (x_L + x_R)/2$;

 if ($f(x_L)f(x_{k+1}) < 0$)

$x_R = x_{k+1}$;

 else

$x_L = x_{k+1}$

 end if;

$k = k + 1$

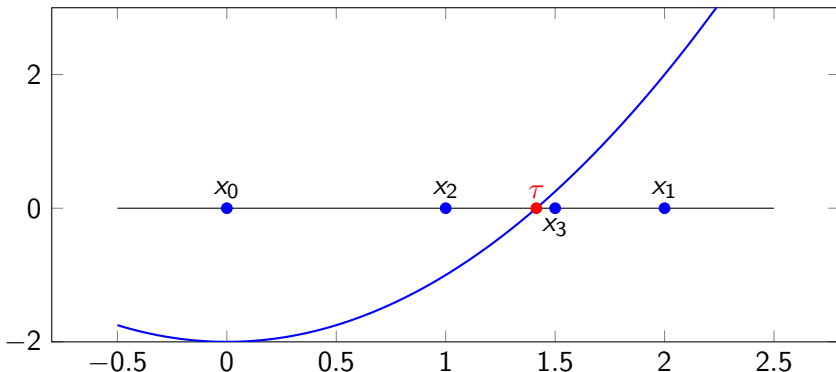
end while;

1. Bisection

Example

Find an estimate for $\sqrt{2}$ that is correct to 6 decimal places.

Solution: Use bisection to solve $f(x) := x^2 - 2 = 0$ on the interval $[0, 2]$.



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k	x_k	$ f(x_k) $	$ \tau - x_k $
0	0.000000	2.00e+00	1.41e+00
1	2.000000	2.00e+00	5.86e-01
2	1.000000	1.00e+00	4.14e-01
3	1.500000	2.50e-01	8.58e-02
4	1.250000	4.38e-01	1.64e-01
5	1.375000	1.09e-01	3.92e-02
6	1.437500	6.64e-02	2.33e-02
7	1.406250	2.25e-02	7.96e-03
8	1.421875	2.17e-02	7.66e-03
9	1.414062	4.27e-04	1.51e-04
$\vdots$	$\vdots$	$\vdots$	$\vdots$
22	1.414214	1.62e-06	5.72e-07
23	1.414214	2.69e-07	9.50e-08

2. The bisection method works

One of the main advantages of the Bisection method is that it will always work, providing only that f is continuous on $[a, b]$, and that the solution exists. Furthermore, we can prove...

Theorem 1.2.1

Let x_k be the k th iteration generated by the Bisection Method. Then

$$|\tau - x_k| \leq \left(\frac{1}{2}\right)^{k-1} |b - a|, \quad \text{for } k = 2, 3, 4, \dots$$

2. The bisection method works

3. Improving upon bisection

A disadvantage of bisection is that it is not particularly efficient. So our next goal will be to derive better methods, particularly the **Secant Method** and **Newton's method**. We also have to come up with some way of expressing what we mean by “**better**”.

4. Exercises

Exercise 1.2.1

Suppose we want to find $\tau \in [a, b]$ such that $f(\tau) = 0$ for some given f , a and b . Write down an estimate for the number of iterations K required by the bisection method to ensure that, for a given ε , we know $|x_k - \tau| \leq \varepsilon$ for all $k \geq K$. In particular, how does this estimate depend on f , a and b ?

Exercise 1.2.2

How many (decimal) digits of accuracy are gained at each step of the bisection method? (If you prefer, how many steps are needed to gain a single (decimal) digit of accuracy?)

4. Exercises

Exercise 1.2.3

Let $f(x) = e^x - 2x - 2$. Show that there is a solution to the problem: *find $\tau \in [0, 2]$ such that $f(\tau) = 0$.*

Taking $x_0 = 0$ and $x_1 = 2$, use 6 steps of the bisection method to estimate τ . You may use a computer program to do this, but please note that in your solution.

Give an upper bound for the error $|\tau - x_6|$.

4. Exercises

Exercise 1.2.4

We wish to estimate $\tau = \sqrt[3]{4}$ numerically by solving $f(x) = 0$ in $[a, b]$ for some suitably chosen f , a and b .

- (i) Suggest suitable choices of f , a , and b for this problem.
- (ii) Show that f has a zero in $[a, b]$.
- (iii) Use 6 steps of the bisection method to estimate $\sqrt[3]{4}$.
You may use a computer program to do this, but please note that in your solution.
- (iv) Use Theorem 7 to give an upper bound for the error $|\tau - x_6|$.